

FCT SERVICE NOTE

SN 06013108a

Class A Service information enclosed.

Date: January 31, 2006

Subject: VCT Gantry RCIB Errors

Effectivity: VCT systems (7.x systems)

Details:

Recent failures of the RCIB communication on the gantry have shown a weakness in understanding of the error messages now displayed with VCT systems. The information below defines the decoding of the RCIB fault error messages with guidelines as to which portion of the chain may be the problem. The RCIB fault is transmitted over redundant paths, a serial and parallel path.

A parallel fault is the same as the serial fault but it shows up everywhere. The same signal drives the parallel and serial fault output drivers. The faults are redundant but the serial fault gives an indication of who drove the fault so that is the focus of the information below.

Serial Fault Troubleshooting

Typical Error message seen:

Host: orp Process: orp_subsys

File: rcib_fault_status.c Line: 189

Exception Class : Secondary Severity : Pri/Soft

File : rcib_fault_status.c Line# : 189

Function : No System Function Reported

Scan Type : None/Unknown Scan: 5/1/0

RCIB Fault Line failure detected.

The status of the fault bus is as follows:

Parallel or Serial Fault line: [1=Fault 0=OK -1=Unknown]

Command: [1=Fault Asserted 0=OK -1=Unknown]

(Note: An unknown status is due to message failure.)

ORP fault status: 0 [1=serial or parallel fault set. 0=OK]

Node	Parallel	Serial	Command
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ORP	-	1	-1
CCB	0	1	0
DCB	0	0	0

Definitions of error message text:

Node: This lists the controller that is reporting.

Parallel: The RCIB has two error paths. One is a path that goes on the CAN bus as a CAN signal. This is the parallel fault path. The controllers will command a fault on both the serial and parallel paths. They are redundant for safety.

A “-” means not read. This should only be indicated on the ORP.

A “1” means the board reads back a fault.

A “0” means the board reads back no fault.

A “-1” means the board has not read the fault status and it is unknown. You probably won't see this.

Serial: This indicates the level sensed on the serial fault path (PSOK signal on the schematics) after the serial fault switch at each node. The serial fault path is basically the power supply looping from the ORP, through the CCB to the DCB. From the DCB it is fed back through the CCB to the ORP and each node has a switch to interrupt the line. See the simplified block diagram.

Command: This indicates whether the firmware on the node has commanded a fault or not. This is the readback of the command, not the status of the actual line.

A “1” means the board has commanded a fault.

A “0” means the board is not commanding a fault.

A “-1” means the command is unknown.

The following table attempts to indicate where there are failures (parts, cables) other than the condition that created a fault command on a node. A board in reset will command a fault.

Reference Error format:

Node	Par	Ser	Command

ORP	-	1	-1
CCB	0	1	0
DCB	0	0	0

Node / Ser		Explanation (reference diagram on Page 5)
ORP	0	Everything is okay
CCB	0	
DCB	0	
ORP	1	The ORP detects a break in the serial line but the CCB and DCB don't. The break is between the sensor on the CCB and the ORP. If the ORP is not commanding an error, the most probable cause is an open cable.
CCB	0	
DCB	0	
ORP	1	The ORP and CCB detect break in the serial line but the DCB doesn't. The break is between the sensor on the DCB and the ORP. If the ORP or CCB are not commanding an error, the most probable causes are an open connection between the CCB and ORP, a bad part on the CCB or a bad connection between the DAS and the CCB. The DAS backplane is also in the path but it is a lower probability of failure. Even lower probability is a bad connector on the DCB-Backplane interface.
CCB	1	
DCB	0	
ORP	1	All nodes detect an error. If the DCB is not commanding an error there is either a bad part on the DCB or the isolated power is not supplied to the DCB. Check your cables between the ORP, Collimator and DAS, the ORP output or the DCB board seating.
CCB	1	
DCB	1	

The following table shows what you should see if the nodes are commanding faults and there are no other part or interconnect failures.

Node	Par	Ser	Command	Explanation
ORP	-	0	0	Everything is okay
CCB	0	0	0	
DCB	0	0	0	
ORP	-	1	1	ORP commands a fault and reads it.
CCB	0	0	0	
DCB	0	0	0	
ORP	-	1	0	The CCB commands a fault and reads it. The ORP also sees the fault line is open.
CCB	0	1	1	
DCB	0	0	0	
ORP	-	1	0	The DCB commands a fault and reads it. The other nodes also see the line is open.
CCB	0	1	0	
DCB	0	1	1	

You can also get faults if one of the relays is shorted and a fault is commanded. You will see this in during a reset or power up condition.

Node	Par	Ser	Command	Explanation
ORP	-	0	0	The DCB commands a fault and reads it. The feedback does not see the fault because the relay on the DCB is shorted.
CCB	0	0	0	
DCB	0	0	1	

The attached diagram also may aid in being able to troubleshoot the fault path. The circuit across the bottom of the page is the serial fault line.

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